Tiago Carneiro Pessoa

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HPC Software Engineer with extensive experience in performance analysis, benchmarking, and optimization of applications on large-scale distributed and heterogeneous systems. Strong background in MPI, OpenMP, GPU programming, and performance-portable HPC software development, with hands-on experience supporting users and projects in deploying, scaling, and optimizing compute-intensive workloads on HPC platforms.

SKILLS

HPC & Programming	C/C++, Python, Chapel, OpenMP, CUDA, HIP, MPI, SIMD
Performance Profiling & Analysis	NVIDIA Nsight, Rocmprof, PERF, Intel VTune, LIKWID
Platforms & Tools: Linux/Unix	Git, Build Systems (CMake, Make), Containers (Docker, Singularity)
Domain Expertise	HPC, distributed systems, heterogeneous computing, performance engineering, combinatorial optimization

LANGUAGES

· Portuguese (Native), English (full professional proficiency), French (professional working proficiency).

EDUCATION

PhD in Computer Science 2013 - 2017
Federal University of Ceará (UFC). Fortaleza - CE, Brazil

Master's Degree in Computer Science 2010 - 2012
State University of Ceará (UECE). Fortaleza - CE, Brazil

Bachelor's Degree in Computer Science 2004 - 2009
State University of Ceará (UECE). Fortaleza - CE, Brazil

AWARDS AND HONORS

Outstanding Paper Award Awarded 2021

The International Conference on High Performance Computing & Simulation - HPCS 2020

WORK EXPERIENCE - FROM PHD

Postdoctoral researcher | Centre INRIA de l'Université de Lille, France Jun 2026 - Present

· Investigate and design heuristic and exact approaches to the qubit allocation problem, in collaboration with PhD students, including mapping and routing. Develop parallel and distributed implementations of these algorithms, with a focus on scalability and productivity.

Researcher | Imec - Leuven, Belgium Sep 2022 - Jan 2025

· Contributed to the design and development of a distributed PGAS-like programming library for novel HPC/AI platforms, together with associated profiling and benchmarking tools.

· Designed and implemented a robust Bash-based distributed process management system to deploy, run, and analyze applications across HPC clusters, supporting performance studies and user workflows.

· Implemented and analyzed representative HPC benchmarks (e.g., FEM, BLAS, HPCG) to study application scalability and performance behavior to validate system and hardware design choices.

Research associate | University of Luxembourg - Belval, Luxembourg Mar 2021 - Sep 2022

· Designed and implemented distributed optimization applications for CPU–GPU heterogeneous systems, focusing on scalability, performance portability, and productivity on large-scale HPC platforms.
· Continued investigating the use of Chapel for productivity and portability on heterogeneous and distributed HPC systems.
· Organized workshops and developed hands-on tutorial material on parallel programming and HPC technologies for students and researchers.

Postdoctoral researcher | INRIA Lille - Nord Europe - Lille, France Nov 2018 - Jun 2020

· Co-created the Chapel-based Optimization Project (ChOP), a recognized initiative contributing to high-performance, distributed Branch-and-Bound algorithms.
· Designed and implemented large-scale distributed applications, focusing on performance portability, scalability, and productivity on heterogeneous HPC systems using high-productivity languages (Chapel, Julia, Python/Numba) and HPC technologies (C/C++, OpenMP, MPI and PGAS).
· Authored multiple papers in top-tier journals and conferences; taught Master's courses and organized workshops on HPC and optimization.

Postdoctoral researcher | IFCE - Fortaleza, Brazil Mar 2018 - Oct 2018

. Supported the integration of GPU acceleration and parallel I/O into computer vision and deep learning applications.
. Used CUDA, OpenCV and Python for GPU-accelerated image processing, inference and performance evaluation.
. Developed tutorial material on Python multiprocessing and provided technical support to students and researchers.

Doctoral Researcher | Federal University of Ceará - Fortaleza, Brazil Mar 2013 - Dec 2017

· Designed and implemented exact parallel and heterogeneous algorithms for combinatorial optimization.
· Achieved significant performance improvements by exploiting GPU acceleration with CUDA Dynamic Parallelism (CDP) and CPU parallelism using SIMD (SEE, AVX), MPI and OpenMP.

Visiting Researcher | INRIA Lille - Nord Europe - Lille, France Sep 2015 - Aug 2016

· As part of my PhD, I researched on parallel optimization algorithms with the Dolphin team, collaborating in an international research environment at INRIA Lille.

SELECTED PUBLICATIONS:

[**1**] Carneiro, T.; Kayraklioglu, E.; Helbecque, G.; Melab, N. Investigating Portability in Chapel for Tree-based Optimization on GPU-powered Clusters. The 30th International European Conference on Parallel and Distributed Computing - Europar 2024. 10.1007/978-3-031-69583-4_27.

[**2**] Carneiro, T.; Melab, N.; Hayashi, A.; Sarkar, V. Towards Chapel-based Exascale Tree Search Algorithms: dealing with multiple GPU accelerators. In: The International Conference on High Performance Computing & Simulation - HPCS 2020 - held in March 2021. **Outstanding Paper Award - HPCS 2020**.

More information about my publication record and citations can be found in my Google Scholar, Orcid (0000-0002-6145-8352) and Web Of Science (J-3061-2019) profiles.